Advanced Electricity & Magnetism Learning Lab

Lab Record & Viva-Voce Booklet

18 Experiments — printable, fill-by-hand observation, calculation, result and viva sheets

Companion to the interactive simulator and online User Guide

Student Name: ____

Roll / Enrolment No.: ____

Course & Section: ____

Academic Session: 2026–27

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How to Use This Booklet

This booklet mirrors the 18 experiments in the online **User Guide** and the interactive simulator. For each experiment: run the simulator panel as instructed in the guide, record your readings directly in the observation table below, plot your graph on the provided 15×15 cm grid (choose your own scale per square), complete the calculation and result sections, work out the maximum permissible error using the least counts of the instruments/settings you used, and prepare written answers to the viva-voce questions before your practical examination. Full theoretical background, governing formulas and step-by-step procedure for every experiment are available in the online guide — this booklet intentionally contains only the record-keeping sheets, not the theory, to keep it compact for printing.

Contents

1. Coulomb's Law & Force Between Point Charges

Aim

To verify the inverse-square dependence of the electrostatic force between two point charges on their separation, and its direct proportionality to the product of the charges, using a torsion-balance style virtual setup.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$F = q_1 q_2 / (4\pi\varepsilon_0\varepsilon_r r^2)$$

F = electrostatic force between the charges (N)
q_1, q_2 = the two point charges (C)
r = separation between the charges (m)
ε_0 = permittivity of free space (8.854×10^{-12} F/m)
ε_r = relative permittivity of the surrounding medium

Labelled diagram of the experimental apparatus:

Observation Table

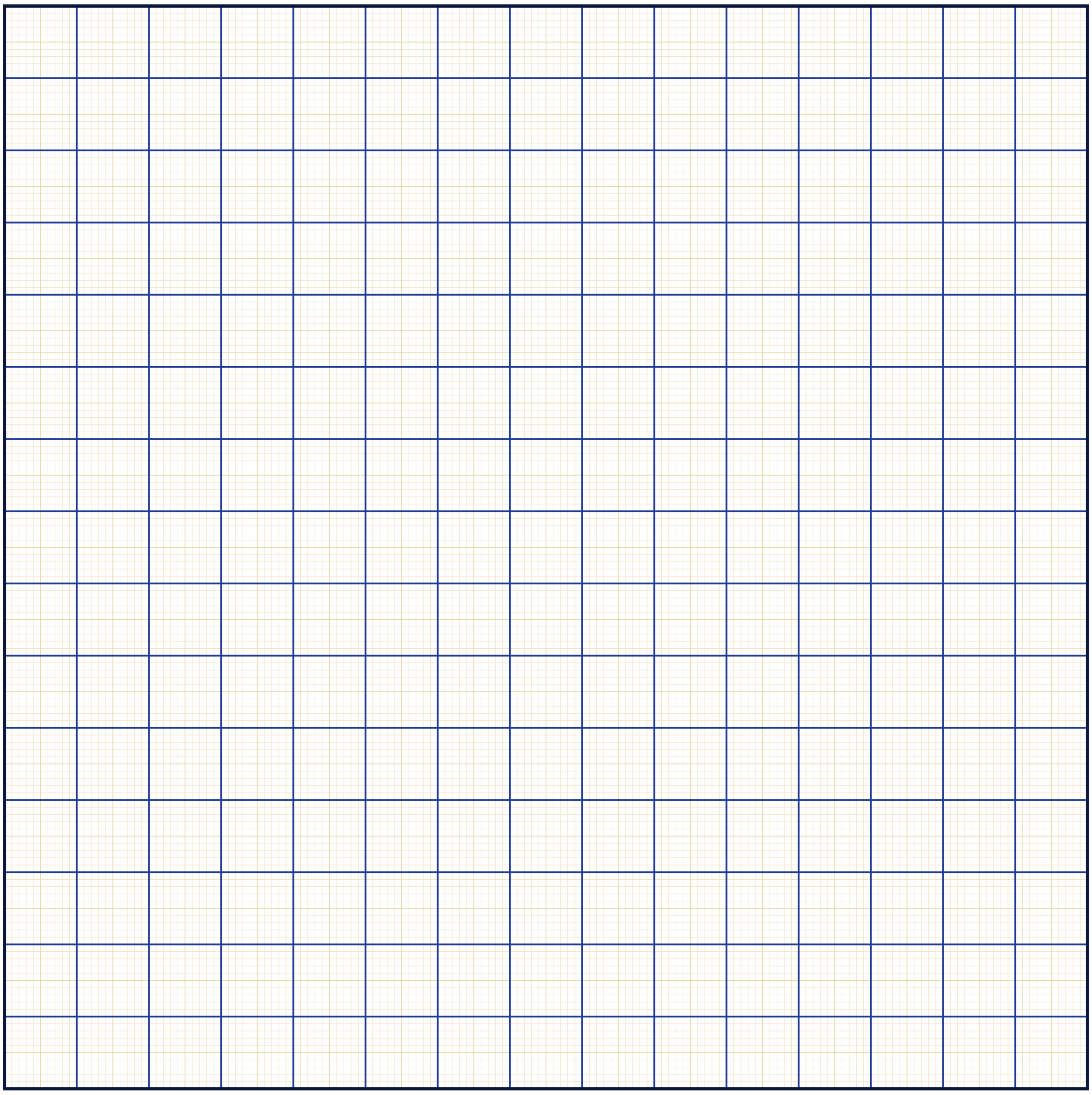
S. No.	Separation r (cm)	q_1 (nC)	q_2 (nC)	Force F ($\times10^{-6}$ N)	log r	log F

S. No.	Separation r (cm)	q_1 (nC)	q_2 (nC)	Force F ($\times 10^{-6}$ N)	log r	log F

Graph

log F vs log r (slope method)

$\log_{10}$ F (F in N)



$\log_{10}$ r (r in m)

Calculation

Result

Quantity	Slope of log F vs log r	Expected value	% deviation

Maximum Permissible Error

Formula: $\Delta F/F = \Delta q_1/q_1 + \Delta q_2/q_2 + 2\Delta r/r$

$F \propto q_1 q_2/r^2$ is a product/quotient of measured quantities, with r entering squared, so its fractional error carries a factor of 2.

Quantity	Measured Value	Least Count (LC)
q■ — Charge 1 (nC)		
q■ — Charge 2 (nC)		
r — Separation (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State Coulomb's law in vector form and explain each term.

2. Why does the electrostatic force obey an inverse-square law rather than, say, an inverse-cube law?

3. How does the force change if the experiment is repeated in a dielectric medium instead of vacuum?

4. Derive the electric field of a point charge starting from Coulomb's law.

5. Compare the magnitude of the electrostatic and gravitational forces between two electrons.

6. What is the principle of superposition, and how does it apply to a system of more than two charges?

2. Electric Field & Potential Due to an Electric Dipole

Aim

To compute and plot the electric field and potential of an electric dipole along its axial and equatorial lines, and to study the angular variation of the field at a fixed distance.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$V = p\cdot\cos\theta / (4\pi\varepsilon_0 r^2),\ E\_axial = 2kp/r^3,\ E\_equatorial = kp/r^3$$

p = dipole moment, $p = q\cdot 2a$ (C·m)
r = distance from the dipole centre (m)
θ = angle from the dipole axis
k = $1/4\pi\varepsilon_0 = 8.988\times10^9$ N·m²/C²

Labelled diagram of the experimental apparatus:

Observation Table

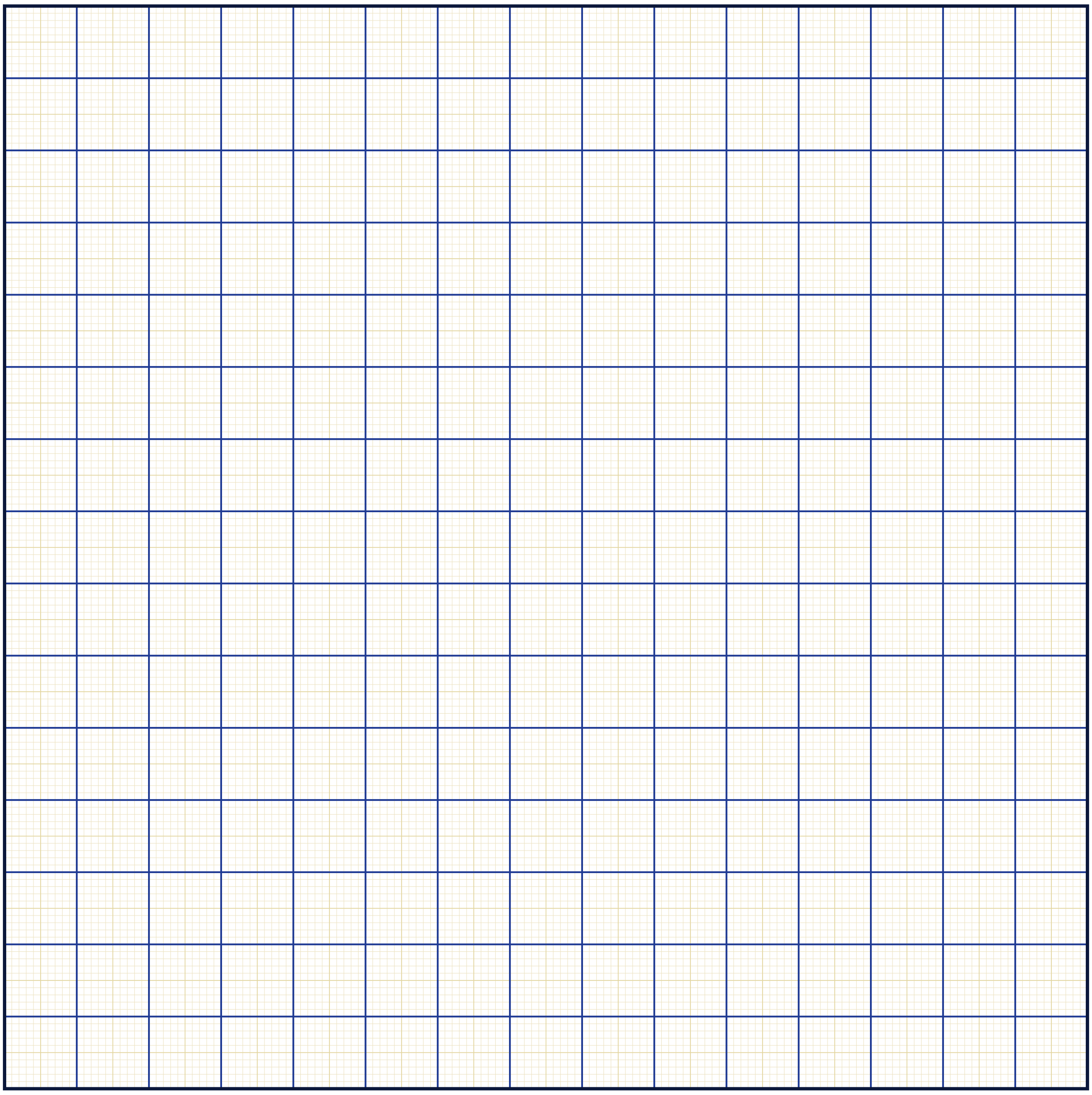
S. No.	θ (°)	r (cm)	Field E (N/C)	Potential V (volt)	log r	log E

S. No.	θ (°)	r (cm)	Field E (N/C)	Potential V (volt)	log r	log E

Graph

Axial Field E vs r (log-log)

log₁₀ E



log₁₀ r

Calculation

Result

Line	Field formula verified	Slope obtained	Expected slope

Maximum Permissible Error

Formula: $\Delta E/E = \Delta q/q + \Delta(2a)/(2a) + 3\Delta r/r$

$E_{axial} = 2kp/r^3$ with $p=q\cdot(2a)$ varies as r^{-3}, so its fractional uncertainty carries a factor of 3 from r.

Quantity	Measured Value	Least Count (LC)
q — Charge magnitude (nC)		
2a — Charge separation (cm)		
r — Probe distance (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the dipole field fall off as $1/r^3$ while a single point charge's field falls off as $1/r^2$?

2. Show that the potential on the equatorial line of a dipole is zero everywhere.

3. What is the torque on a dipole placed in a uniform external field, and at what orientation is it maximum?

4. Define dipole moment as a vector and state its SI unit.

5. How would the field pattern change for a linear quadrupole (two dipoles placed back-to-back)?

6. Explain the physical origin of polar molecules' permanent dipole moment (e.g. in water).

3. Gauss's Law — Electric Flux Through Closed Surfaces

Aim

To verify Gauss's law by computing the net electric flux through virtual closed Gaussian surfaces (sphere, cube, cylinder) enclosing a known charge, and to relate the flux to the field of standard symmetric charge distributions.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\Phi_E = \oint E \cdot dA = Q_{enc} / \varepsilon_0$$

Φ_E = net electric flux through the closed surface ($N \cdot m^2/C$)
Q_{enc} = total charge enclosed by the surface (C)
ε_0 = permittivity of free space

Labelled diagram of the experimental apparatus:

Observation Table

S. No.	Surface shape	Q_{enc} (nC)	Q_{out} (nC)	Computed flux Φ ($N \cdot m^2/C$)	Q_{enc}/ε_0 (theory)

S. No.	Surface shape	Q_{enc} (nC)	Q_{out} (nC)	Computed flux Φ ($N \cdot m^2/C$)	Q_{enc}/ε_0 (theory)

Graph

Field of a Charged Sphere: E vs r

E (relative units)

r (cm)

Calculation

Result

Case	Flux measured	Flux predicted (Q_{enc}/ε_0)	% deviation

Maximum Permissible Error

Formula: $\Delta\Phi/\Phi = \Delta Q_{enc}/Q_{enc}$

Since $\Phi = Q_{enc}/\varepsilon_0$ with ε_0 an exact constant, the fractional error in flux equals the fractional error in the enclosed charge measurement.

Quantity	Measured Value	Least Count (LC)
Q_{enc} — Enclosed charge (nC)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State Gauss's law and explain why it is a direct consequence of Coulomb's law and the inverse-square dependence.

2. Why is Gauss's law most useful only for highly symmetric charge distributions?

3. Show that the field inside a uniformly charged spherical shell is exactly zero.

4. Derive the field of an infinite charged sheet using Gauss's law with a pillbox surface.

5. If a charge lies exactly on the Gaussian surface, how should it be treated?

6. How does Gauss's law extend to Gauss's law for magnetism ($\nabla\cdot B = 0$), and what does that imply physically?

4. Capacitance of a Parallel-Plate Capacitor (With & Without Dielectric)

Aim

To study the dependence of a parallel-plate capacitor's capacitance on plate area, plate separation, and an inserted dielectric slab, and to determine the dielectric constant of an unknown material.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$C = \varepsilon_r \varepsilon_0 A/d$$

C = capacitance (F)
εr = relative permittivity (dielectric constant) of the medium
A = overlap area of the plates (m^2)
d = plate separation (m)

Labelled diagram of the experimental apparatus:

Observation Table

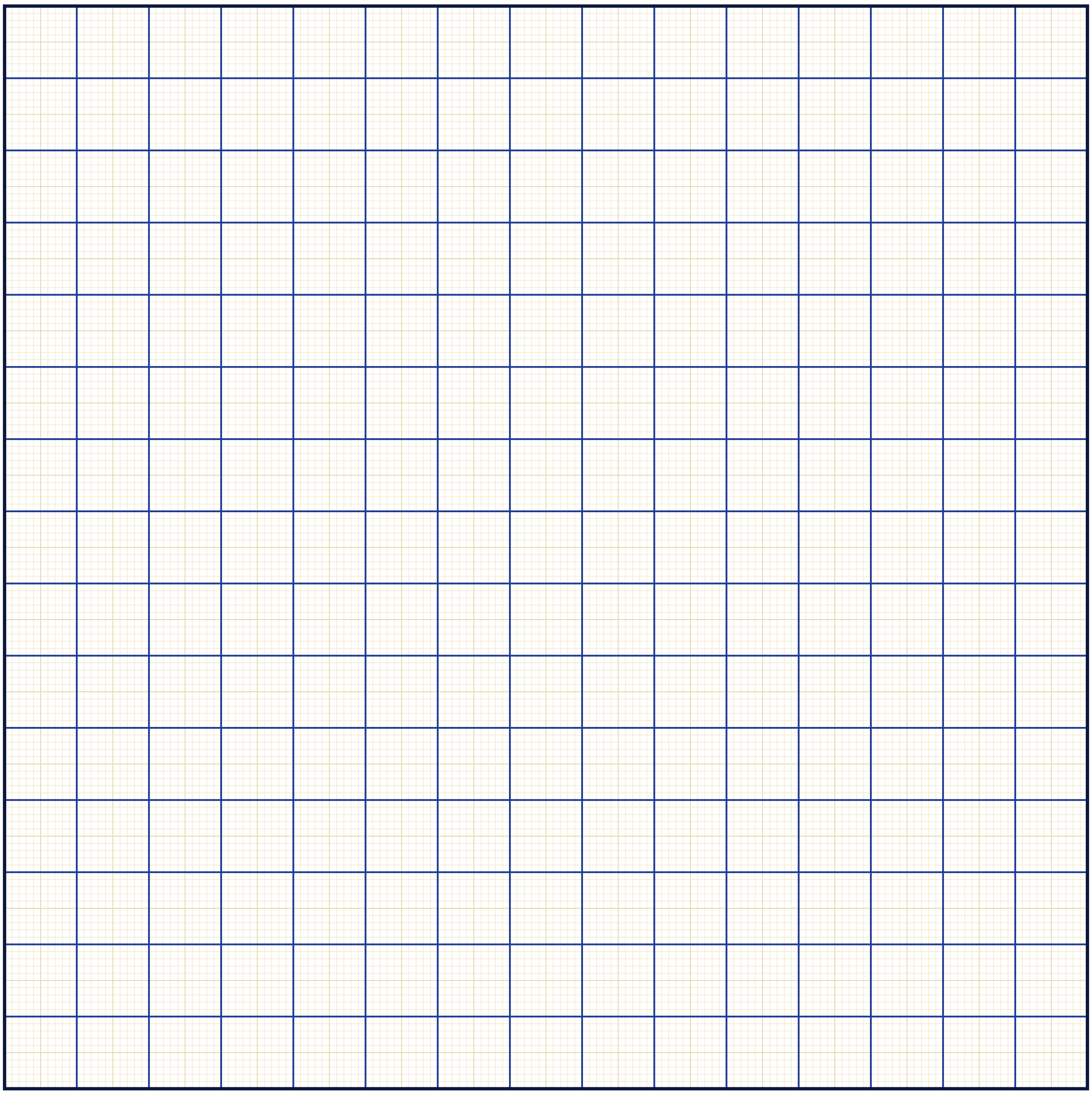
S. No.	d (mm)	A (cm^2)	ε_r	Capacitance C (pF)	1/d (mm^{-1})

S. No.	d (mm)	A (cm^2)	εr	Capacitance C (pF)	1/d (mm^{-1})

Graph

Capacitance vs 1/d

C (pF)



1/d (mm^{-1})

Calculation

Result

Dielectric material	Measured $\varepsilon r = C/C_0$	Known/standard εr	% deviation

Maximum Permissible Error

Formula: $\Delta C/C = \Delta\varepsilon r/\varepsilon r + \Delta A/A + \Delta d/d$

$C = \varepsilon r \varepsilon_0 A/d$ is a product/quotient of three measured quantities, so fractional errors add.

Quantity	Measured Value	Least Count (LC)
εr — Dielectric constant		
A — Area (cm^2)		
d — Separation (mm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the capacitance of a parallel-plate capacitor from first principles using Gauss's law.
2. Why does inserting a dielectric always increase the capacitance, never decrease it?
3. Distinguish between the behaviour of a dielectric slab at constant charge versus constant voltage.
4. Define dielectric constant (relative permittivity) and polarization P; how are they related?
5. What is dielectric breakdown, and what determines a material's dielectric strength?

6. How would you find the capacitance of a capacitor with the gap only half-filled by a dielectric slab parallel to the plates?

5. Charge Distribution on Conductors & the Van de Graaff Generator

Aim

To study how charge distributes itself over the surface of conductors of different shapes at electrostatic equilibrium, and to understand the principle of the Van de Graaff generator for building up very high potentials.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$V_{dome} = Q / (4\pi\varepsilon_0 R), \quad E_{surface} = \sigma/\varepsilon_0$$

V_dome = potential of the charged spherical dome (V)
Q = total charge delivered to the dome (C)
R = radius of the dome (m)
σ = local surface charge density (C/m^2)

Labelled diagram of the experimental apparatus:

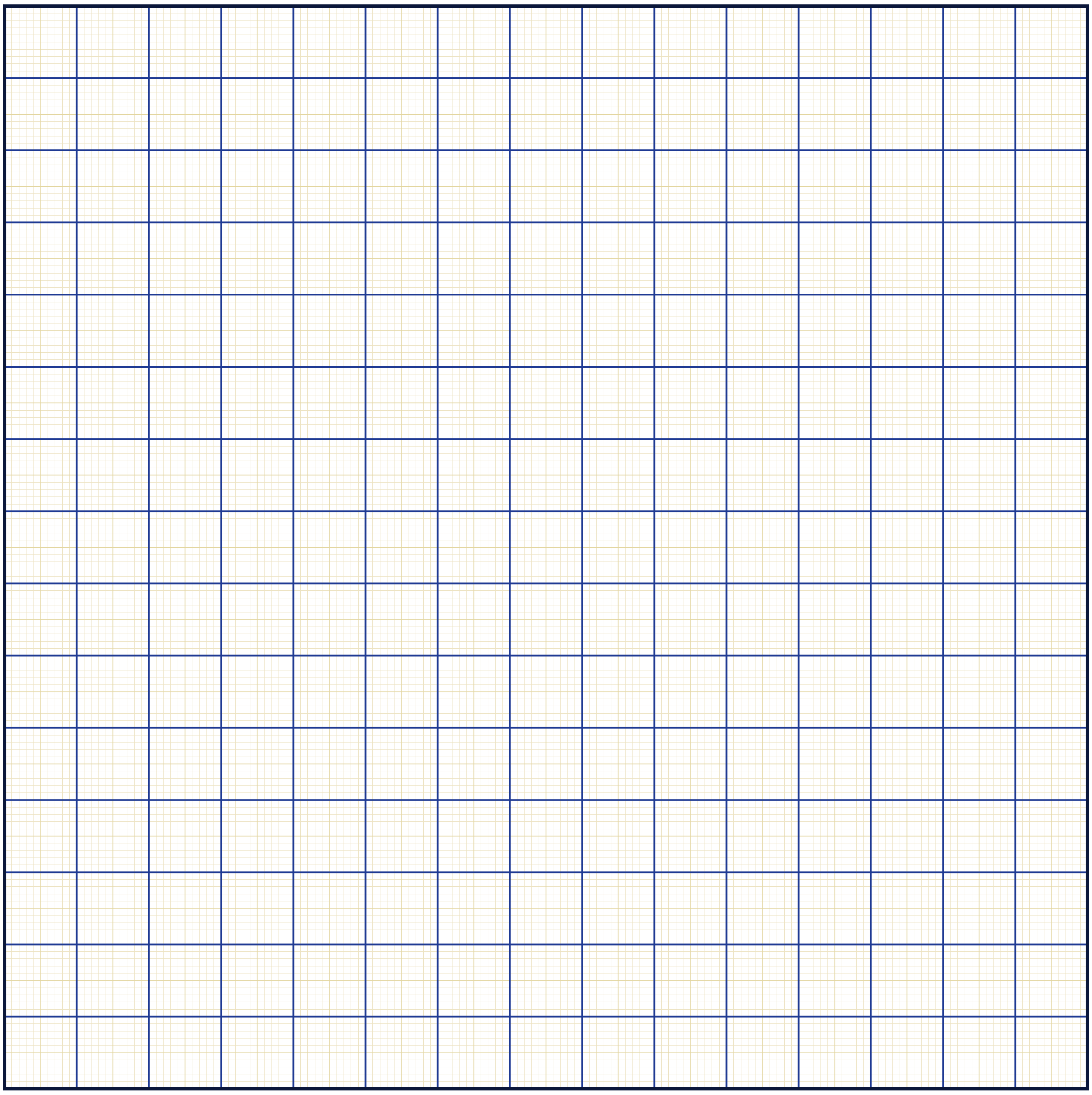
Observation Table

S. No.	Charging time t (s)	Rate (nC/s)	Total charge Q (nC)	Dome radius R (cm)	Dome potential V (kV)

S. No.	Charging time t (s)	Rate (nC/s)	Total charge Q (nC)	Dome radius R (cm)	Dome potential V (kV)

Graph

Dome Potential vs Charging Time



V (kV)

t (s)

Calculation

Result

Quantity studied	Observed trend	Theoretical prediction	Agreement

Maximum Permissible Error

Formula: $\Delta V/V = \Delta rate/rate + \Delta t/t + \Delta R/R$

$V = kQ/R$ with $Q = rate \times t$ (charging rate times time), so fractional errors in rate, time and R all add.

Quantity	Measured Value	Least Count (LC)
rate — Belt charging rate (nC/s)		
t — Charging time (s)		
R — Dome radius (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why must the electric field inside a conductor in electrostatic equilibrium be zero?

2. Explain why charge density is greatest at the sharpest points of a charged conductor.

3. How does a Van de Graaff generator manage to raise the dome to a very high potential regardless of the charge already present on it?

4. What is corona discharge and why does it limit the maximum achievable potential?

5. State the boundary condition on E just outside a charged conductor's surface in terms of σ.

6. Why is the interior of a hollow charged conductor field-free even if the shell is not spherical (Faraday cage)?

6. Biot–Savart Law — Magnetic Field Due to a Current-Carrying Wire

Aim

To compute the magnetic field due to a finite straight current-carrying wire using the Biot–Savart law, verify the perpendicular-distance dependence, and recover Ampere's infinite-wire result as a limiting case.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$B = (\mu_0 I / 4\pi a)(\sin\alpha_1 + \sin\alpha_2)$$

B = magnetic field at the probe point (T)
I = current in the wire (A)
a = perpendicular distance from the wire (m)
α_1, α_2 = angles subtended by the wire's two ends at the field point
μ_0 = permeability of free space ($4\pi\times10^{-7}$ T·m/A)

Labelled diagram of the experimental apparatus:

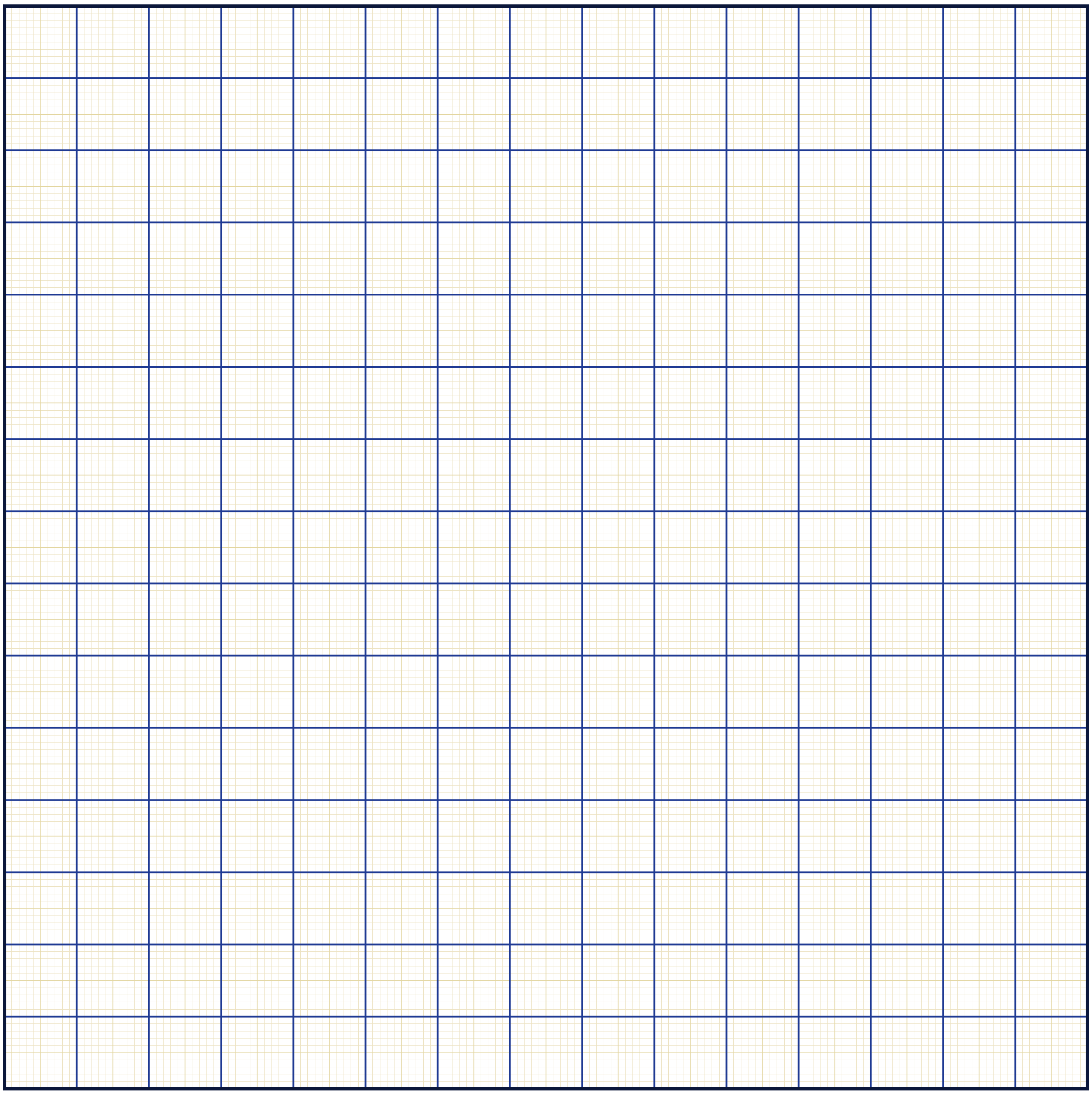
Observation Table

S. No.	Current I (A)	Distance a (cm)	Wire length L (cm)	α_1 (°)	α_2 (°)	Field B ($\times10^{-5}$ T)

S. No.	Current I (A)	Distance a (cm)	Wire length L (cm)	α_1 (°)	α_2 (°)	Field B ($\times 10^{-5}$ T)

Graph

B vs 1/a at fixed I, L



B ($\times 10^{-5}$ T)

1/a (cm^{-1})

Calculation

Result

Wire length L	B (finite wire)	B (infinite-wire formula)	% difference

Maximum Permissible Error

Formula: $\Delta B/B = \Delta I/I + \Delta a/a$

For the infinite-wire limit $B = \mu_0 I/2\pi a$, a simple quotient, so fractional errors in I and a add.

Quantity	Measured Value	Least Count (LC)
I — Current (A)		
a — Distance (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State the Biot–Savart law in vector form and compare its structure with Coulomb's law.

2. Derive the infinite-wire field formula $B = \mu_0 I/2\pi a$ as a limiting case of the finite-wire result.

3. In which direction does B point around a current-carrying wire, and which rule fixes this?

4. Why is the field due to a straight wire independent of its cross-sectional shape (for a thin wire)?

5. How would the Biot–Savart integral change for a wire bent into an L-shape?

6. Compare and contrast the Biot–Savart law with Ampere's circuital law — when is each more convenient to use?

7. Magnetic Field on the Axis of a Circular Coil & Helmholtz Coils

Aim

To study the magnetic field along the axis of a single circular current loop, locate the point of maximum field, and demonstrate how a Helmholtz pair produces a highly uniform field at its centre.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$B(x) = \mu_0 N I R^2 / [2(R^2 + x^2)^{3/2}]$$

B(x) = axial field at distance x from the coil centre (T)
N = number of turns
I = current (A)
R = coil radius (m)
x = axial distance from the coil's centre (m)

Labelled diagram of the experimental apparatus:

Observation Table

S. No.	x (cm)	R (cm)	N	I (A)	Field B ($\times 10^{-4}$ T)

S. No.	x (cm)	R (cm)	N	I (A)	Field B ($\times 10^{-4}$ T)

Graph

B(x) Along the Coil Axis

B ($\times 10^{-4}$ T)

x (cm)

Calculation

Result

Configuration	Field uniformity ($\Delta B/B$ over ±2 cm)	Optimum separation d	Remarks

Maximum Permissible Error

Formula: $\Delta B_0/B_0 = \Delta N/N + \Delta I/I + \Delta R/R$

$B_0 = \mu_0 NI/2R$ at the coil centre is a simple product/quotient, so fractional errors add.

Quantity	Measured Value	Least Count (LC)
N — Turns		
I — Current (A)		
R — Radius (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the on-axis field of a circular current loop starting from the Biot–Savart law.

2. At what axial distance does B(x) fall to half its central value, for a single coil?

3. Why is the Helmholtz coil separation chosen equal to the coil radius?

4. What physical quantity being zero at the midpoint makes the Helmholtz field so uniform there?

5. How would you use a Helmholtz pair to cancel the Earth's magnetic field in a sensitive experiment?

6. Compare the on-axis field of a coil with that of a long solenoid — under what condition does one reduce to the other?

8. Ampere's Circuital Law — Field Inside a Solenoid & a Toroid

Aim

To verify Ampere's circuital law by computing the magnetic field inside a long solenoid and a toroid, and to study the field's uniformity and its dependence on turns density and current.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

Solenoid: $B = \mu_0 n I$ | Toroid: $B(r) = \mu_0 N I / (2\pi r)$

B = magnetic field (T)
n = turns per unit length of the solenoid (turns/m)
N = total turns of the toroid
I = current (A)
r = radial distance from the toroid's axis, within the core (m)

Labelled diagram of the experimental apparatus:

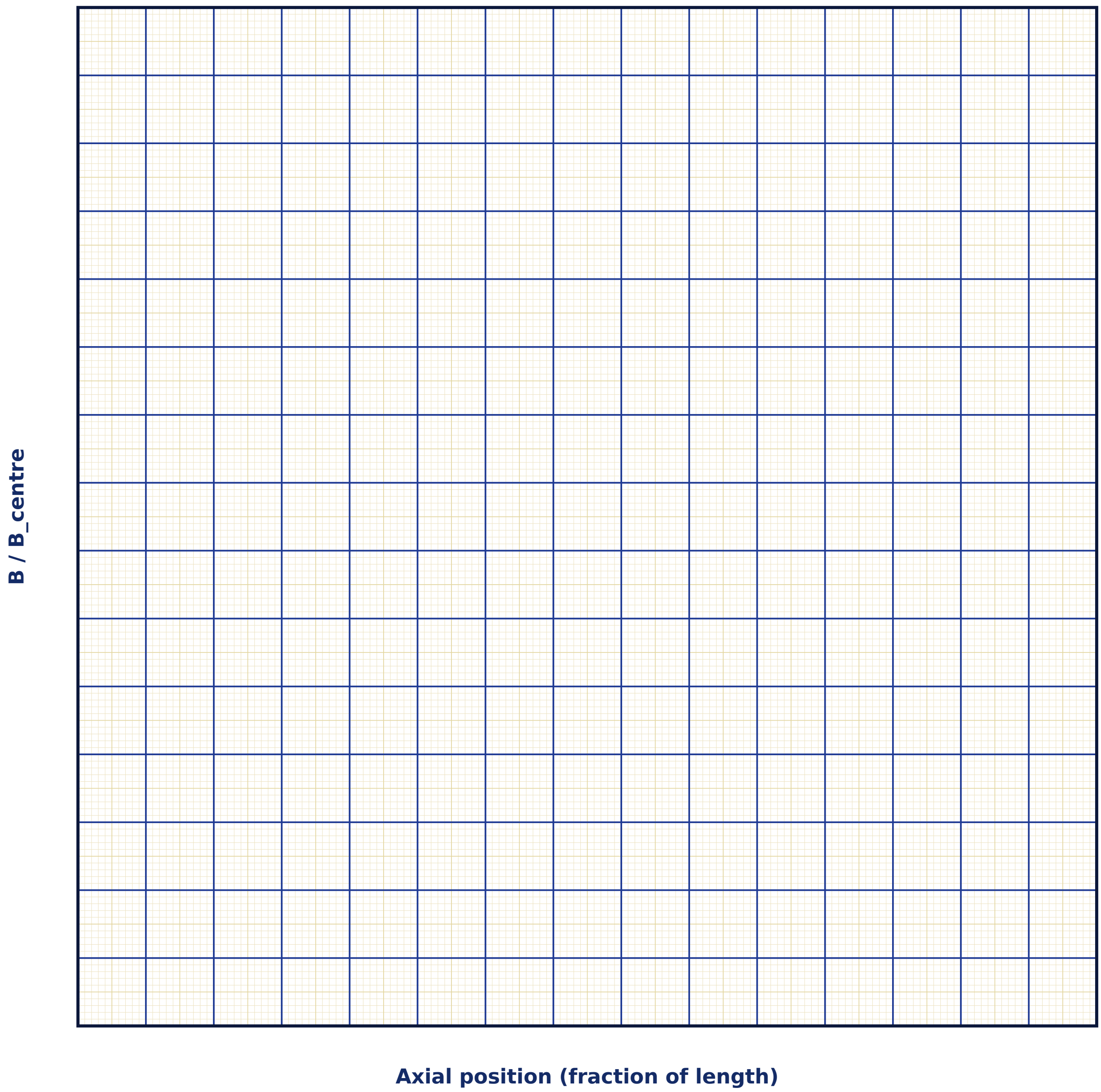
Observation Table

S. No.	Configuration	n or N	Current I (A)	Probe position	Field B ($\times 10^{-3}$ T)

S. No.	Configuration	n or N	Current I (A)	Probe position	Field B ($\times 10^{-3}$ T)

Graph

Solenoid Field Along the Axis (end effects)



Calculation

Result

Configuration	Field uniformity	Verified law	Remarks

Maximum Permissible Error

Formula: $\Delta B/B = \Delta n/n + \Delta I/I$

$B = \mu_0 nI$ is a simple product, so fractional errors in n and I add directly.

Quantity	Measured Value	Least Count (LC)
n — Turns density (turns/cm)		
I — Current (A)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State Ampere's circuital law and explain the choice of Amperian loop used to derive the solenoid's field.

2. Why is the field of an ideal solenoid exactly zero outside the winding, but not exactly zero for a real, finite solenoid?

3. Derive the toroid's field formula using a circular Amperian loop.

4. Why is a toroid's field not perfectly uniform across its cross-section, unlike an idealized long solenoid?

5. Under what conditions does Ampere's law alone suffice to determine B, without needing the full Biot–Savart integral?

6. How is Ampere's law modified for time-varying fields (Maxwell's displacement-current correction)?

9. Force Between Two Parallel Current-Carrying Conductors

Aim

To compute the force per unit length between two long parallel current-carrying wires, verify its dependence on the currents and their separation, and relate it to the SI definition of the ampere.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$F/L = \mu_0 I_1 I_2 / (2\pi d)$$

F/L = force per unit length between the wires (N/m)
I_1, I_2 = currents in the two wires (A)
d = separation between the wires (m)
μ_0 = permeability of free space

Labelled diagram of the experimental apparatus:

Observation Table

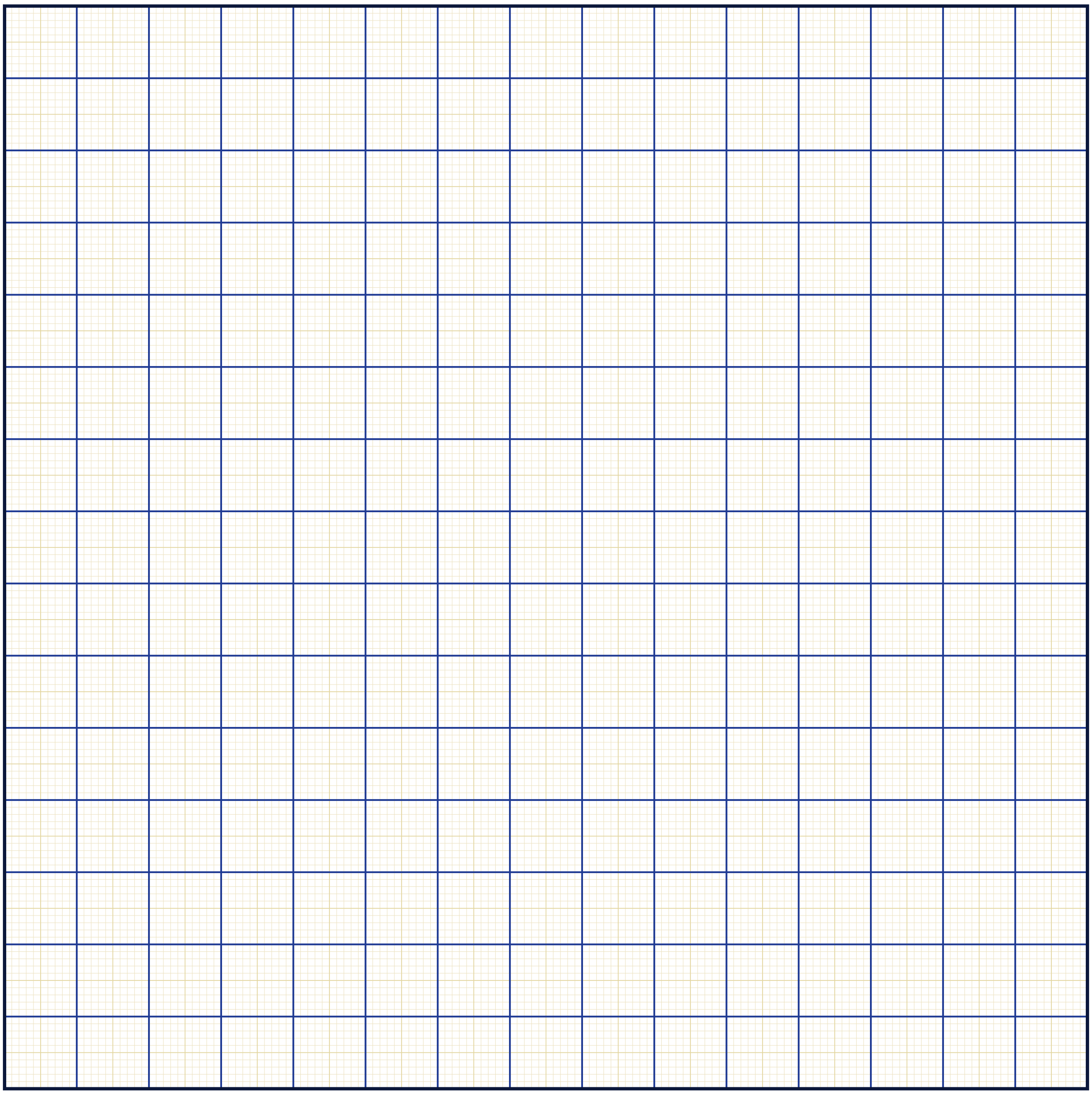
S. No.	I_1 (A)	I_2 (A)	Separation d (cm)	Direction (same/opp.)	Force/length ($\times 10^{-5}$ N/m)

S. No.	I_1 (A)	I_2 (A)	Separation d (cm)	Direction (same/opp.)	Force/length ($\times 10^{-5}$ N/m)

Graph

Force per Unit Length vs 1/d

F/L ($\times 10^{-5}$ N/m)



1/d (cm^{-1})

Calculation

Result

Configuration	F/L measured	F/L predicted	% deviation

Maximum Permissible Error

Formula: $\Delta(F/L)/(F/L) = \Delta I_1/I_1 + \Delta I_2/I_2 + \Delta d/d$

$F/L = \mu_0 I_1 I_2/2\pi d$ is a product/quotient of three measured quantities.

Quantity	Measured Value	Least Count (LC)
I■ — Current 1 (A)		
I■ — Current 2 (A)		
d — Separation (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the force per unit length between two parallel current-carrying wires using the Biot–Savart/Ampere field of one wire acting on the other.

2. Why do parallel currents in the same direction attract while antiparallel currents repel?

3. Explain the historical (pre-2019) SI definition of the ampere in terms of this force.

4. How does the modern (2019 revision) SI definition of the ampere differ, and why was it changed?

5. If both currents are doubled, by what factor does the force per unit length change?

6. How would you measure this force experimentally using a current balance?

10. Faraday's Law of Electromagnetic Induction

Aim

To verify Faraday's law of electromagnetic induction by studying the EMF induced in a coil due to a changing magnetic flux, and to confirm Lenz's law for the direction of the induced current.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\varepsilon = -N \, d\Phi_B/dt$$

ε = induced EMF (V)
N = number of turns in the coil
Φ_B = magnetic flux linking one turn (Wb)
t = time (s)

Labelled diagram of the experimental apparatus:

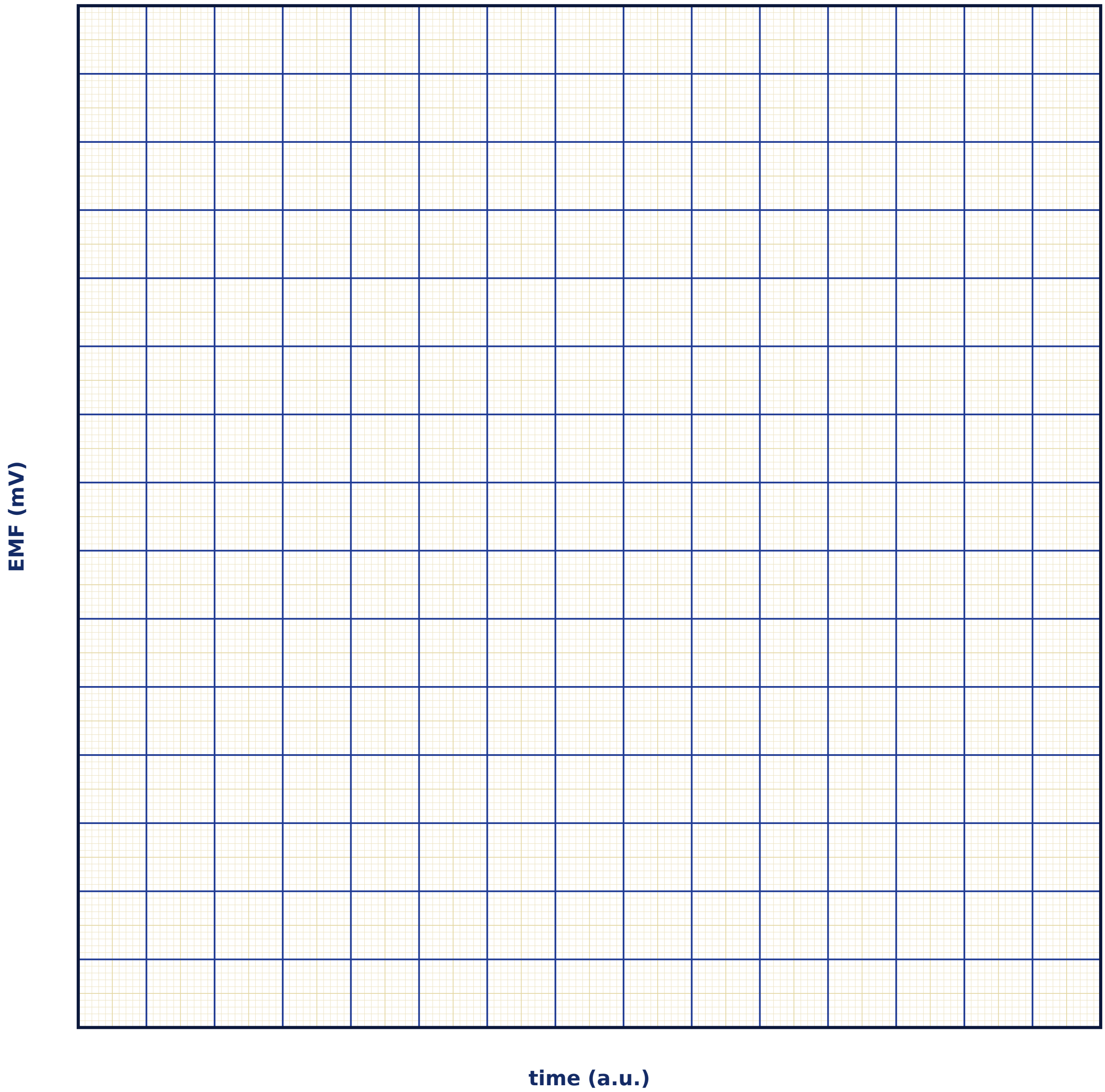
Observation Table

S. No.	Turns N	Magnet speed v (cm/s)	Peak field B_max (mT)	Peak EMF ε (mV)	Direction (approach/withdraw)

S. No.	Turns N	Magnet speed v (cm/s)	Peak field B_max (mT)	Peak EMF ε (mV)	Direction (approach/withdraw)

Graph

Induced EMF vs Time (magnet passing through coil)



Calculation

Result

Case	EMF observed	EMF predicted (−N dΦ/dt)	Lenz's law direction verified?

Maximum Permissible Error

Formula: $\Delta\varepsilon/\varepsilon = \Delta N/N + \Delta B_{max}/B_{max} + \Delta v/v$

For a magnet swept through at speed v, peak EMF scales with N, the field's peak value and its rate of change ($\propto v$), so fractional errors add.

Quantity	Measured Value	Least Count (LC)
N — Turns		
B_max — Peak field (mT)		
v — Speed (cm/s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. State Faraday's law and explain the physical meaning of the negative sign (Lenz's law).

2. Why is no EMF induced by a strong but unchanging magnetic field through a stationary coil?

3. Derive the motional EMF formula $\varepsilon = BLv$ from Faraday's law for a rod sliding on rails in a magnetic field.

4. Explain how Lenz's law is a direct consequence of energy conservation.

5. How does eddy-current braking make use of Faraday's and Lenz's laws?

6. What is self-inductance, and how does it relate to Faraday's law applied to a coil's own changing current?

11. Motional EMF & Lenz's Law on a Sliding Rod

Aim

To study the motional EMF generated by a conducting rod sliding on rails in a uniform magnetic field, verify $\varepsilon = BLv$, and observe the retarding (Lenz's-law) force that opposes the rod's motion when the circuit is closed.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\varepsilon = BLv,\ I = \varepsilon/R,\ F_{retard} = BIL = B^2L^2v/R$$

ε = motional EMF (V)
B = magnetic field (T)
L = rail separation / rod length (m)
v = rod velocity (m/s)
R = circuit resistance (Ω)
F_retard = retarding force on the rod (N)

Labelled diagram of the experimental apparatus:

Observation Table

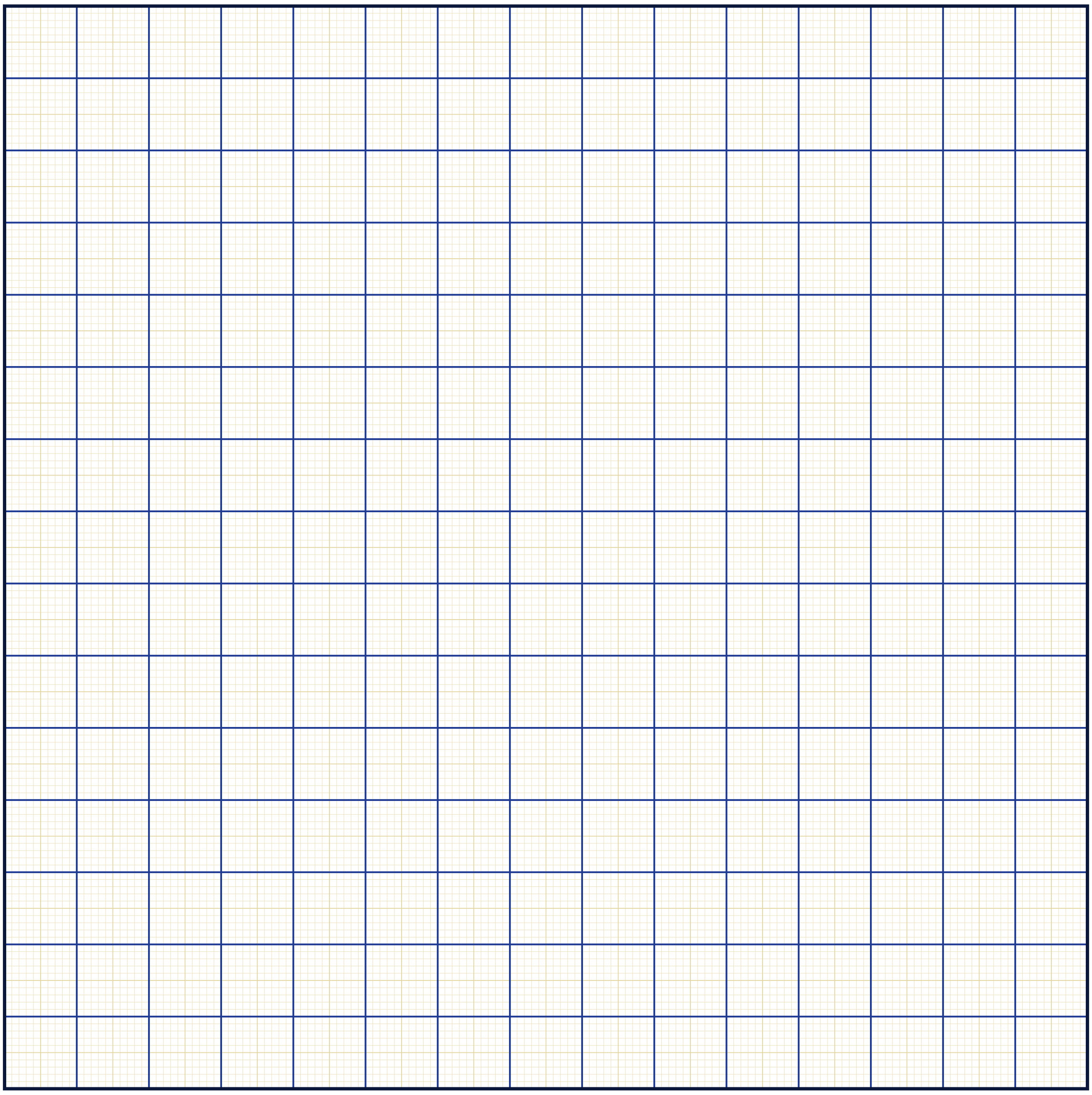
S. No.	B (mT)	L (cm)	v (m/s)	R (Ω)	EMF ε (mV)	Current I (mA)	Retarding force F ($\times 10^{-4}$ N)

S. No.	B (mT)	L (cm)	v (m/s)	R (Ω)	EMF ε (mV)	Current I (mA)	Retarding force F ($\times 10^{-4}$ N)

Graph

Motional EMF vs Rod Velocity

ε (mV)



v (m/s)

Calculation

Result

Quantity	Measured value	Predicted value	% deviation

Maximum Permissible Error

Formula: $\Delta\varepsilon/\varepsilon = \Delta B/B + \Delta L/L + \Delta v/v$

$\varepsilon = BLv$ is a simple product of three measured quantities.

Quantity	Measured Value	Least Count (LC)
B — Field (mT)		
L — Rail separation (cm)		
v — Velocity (m/s)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the motional EMF $\varepsilon = BLv$ from the magnetic force on free charges in the moving rod.

2. Show that the same result follows from Faraday's law applied to the changing enclosed area.

3. Why does the rod experience a retarding force only when the circuit is closed?

4. Show that the electrical power dissipated equals the mechanical power delivered to keep the rod moving at constant velocity.

5. How would the terminal (constant) velocity of a freely falling conducting rod on vertical rails in a field be determined?

6. Explain the connection between this experiment and the working principle of an AC generator.

12. RC and RL Transient Circuits — Charging & Discharging

Aim

To study the exponential charging and discharging behaviour of RC and RL circuits, measure the time constant in each case, and verify its dependence on the circuit parameters.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

RC: $V_C(t) = \varepsilon_0(1-e^{-t/\tau})$, $\tau=RC$ | RL: $I(t) = (\varepsilon_0/R)(1-e^{-t/\tau})$, $\tau=L/R$

τ = time constant of the circuit (s)
R = resistance (Ω)
C = capacitance (F)
L = inductance (H)
ε_0 = source EMF (V)

Labelled diagram of the experimental apparatus:

Observation Table

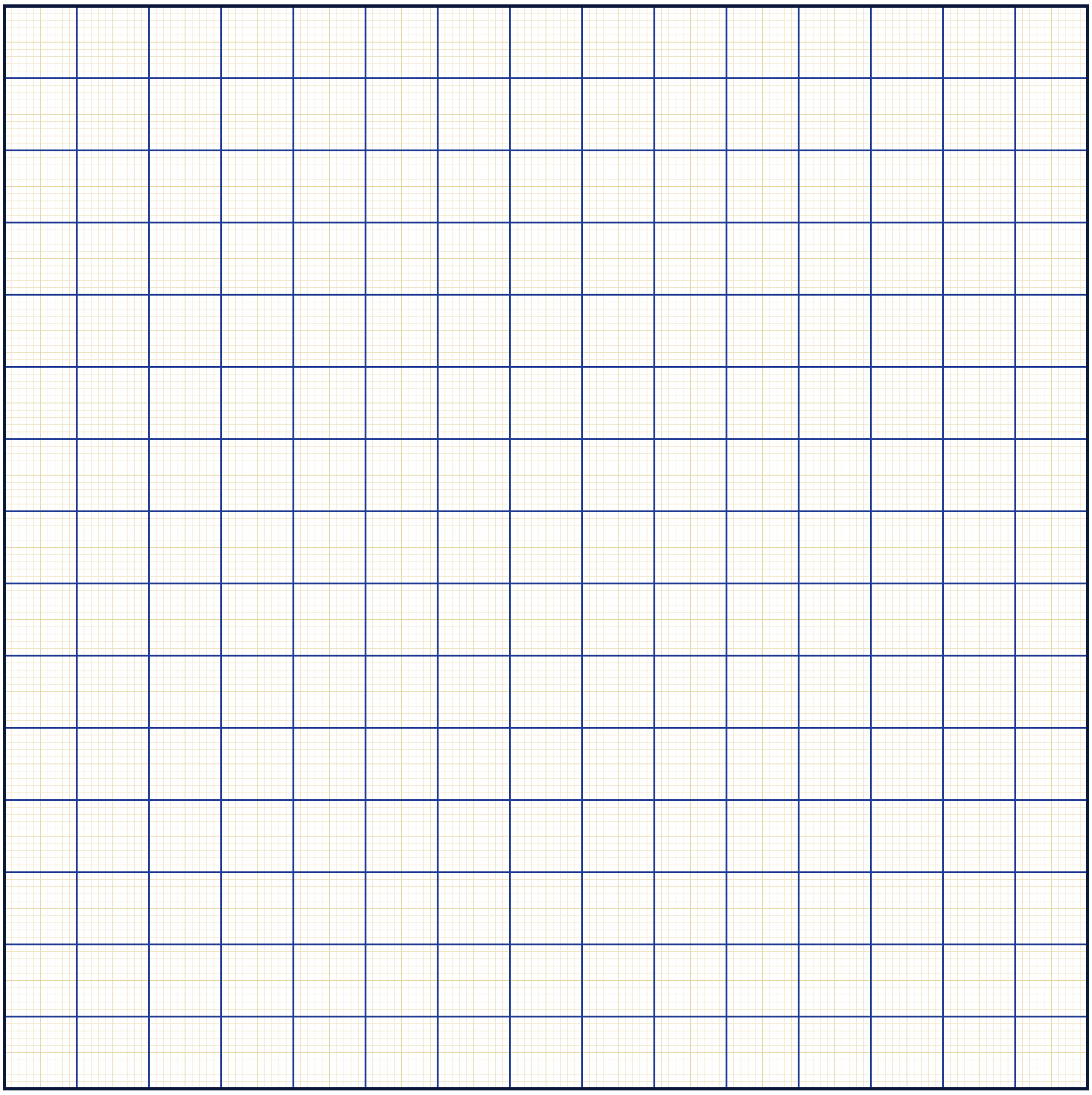
S. No.	Circuit type	R (kΩ)	C (μF) or L (H)	Measured τ (ms)	Calculated τ = RC or L/R (ms)

S. No.	Circuit type	R (kΩ)	C (μF) or L (H)	Measured τ (ms)	Calculated τ = RC or L/R (ms)

Graph

Capacitor Charging Curve

V_C/ε₀



t/τ

Calculation

Result

Circuit	τ measured	τ calculated	% deviation

Maximum Permissible Error

Formula: $\Delta\tau/\tau = \Delta R/R + \Delta C/C$ (RC circuit)

$\tau = RC$ is a simple product, so fractional errors in R and C add directly.

Quantity	Measured Value	Least Count (LC)
R — Resistance (kΩ)		
C — Capacitance (µF)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the charging equation of an RC circuit from Kirchhoff's voltage law.

2. What is the physical significance of the time constant τ, and how is it read off an exponential curve graphically?

3. Why do the RC-charging and RL-growth curves have the same mathematical form?

4. How much energy is ultimately stored in the capacitor after full charging, and how does this compare with the energy dissipated in R during charging?

5. What happens to the time constant if R is doubled while C is halved?

6. Explain the duality between RC and RL transient circuits in terms of the roles of voltage and current.

13. LCR Series Resonance & Quality Factor (Q)

Aim

To study the resonance behaviour of a series LCR circuit driven by an AC source, locate the resonant frequency, and determine the quality factor from the sharpness of the resonance curve.

Date performed: ___________________ Simulator panel used: ____________________________

Formula Used

$$\omega_0 = 1/\sqrt{LC}, \quad Q = \omega_0 L/R = (1/R)\sqrt{L/C}$$

ω_0 = resonant angular frequency (rad/s)
L = inductance (H)
C = capacitance (F)
R = resistance (Ω)
Q = quality factor (dimensionless)

Labelled diagram of the experimental apparatus:

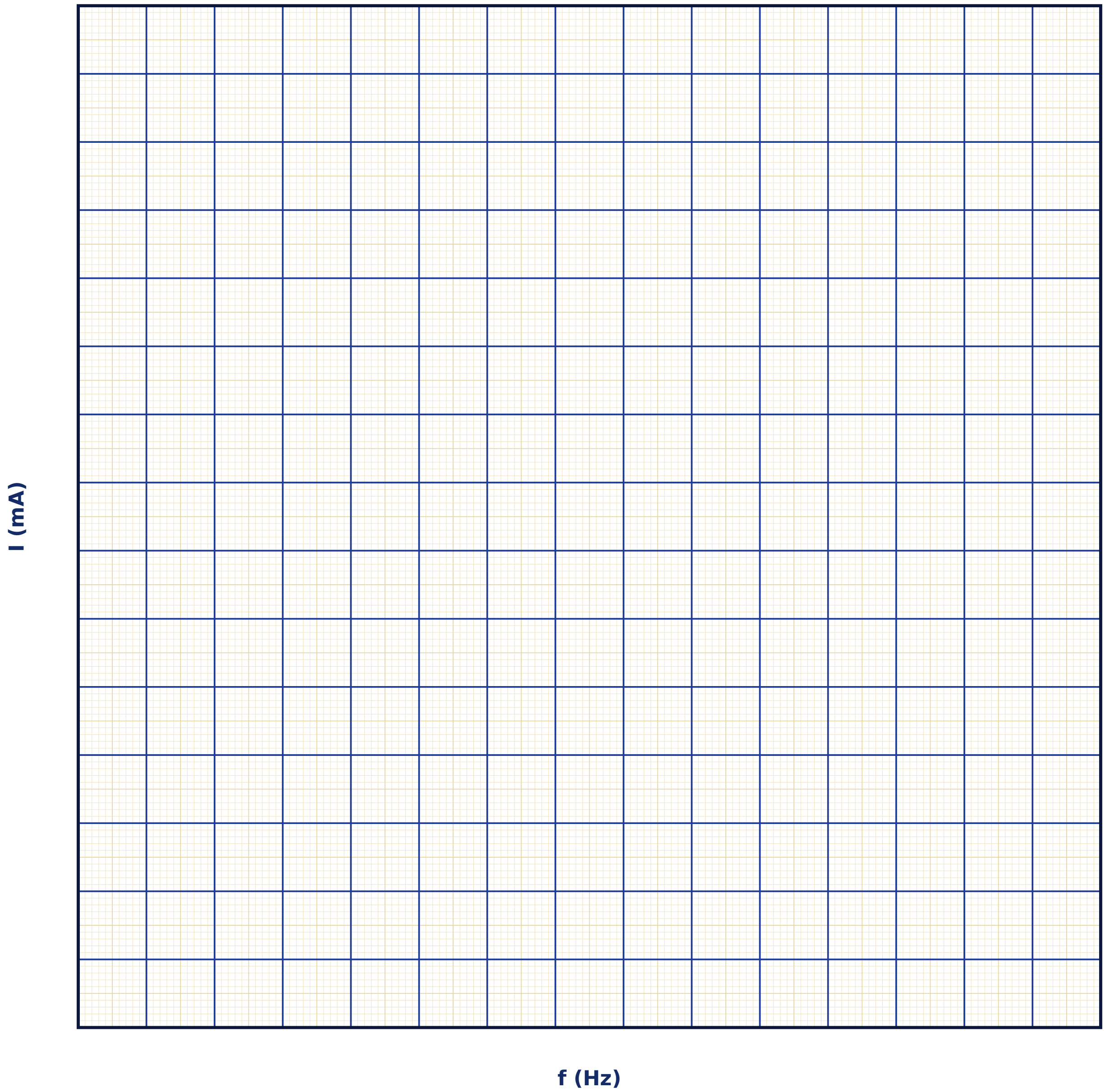
Observation Table

S. No.	R (Ω)	L (mH)	C (μF)	Frequency f (Hz)	Current amplitude I (mA)

S. No.	R (Ω)	L (mH)	C (μF)	Frequency f (Hz)	Current amplitude I (mA)

Graph

Resonance Curve: Current vs Frequency



Calculation

Result

Quantity	Measured value	Calculated value	% deviation

Maximum Permissible Error

Formula: $\Delta f_0/f_0 = \frac{1}{2}(\Delta L/L + \Delta C/C)$

$f_0 = 1/(2\pi\sqrt{LC})$ involves a square root of the product LC, halving the summed fractional errors.

Quantity	Measured Value	Least Count (LC)
L — Inductance (mH)		
C — Capacitance (µF)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the resonant frequency of a series LCR circuit from its impedance expression.

2. Define quality factor Q in at least two equivalent ways and show they are consistent.

3. Why can the voltage across the inductor or capacitor individually exceed the applied source voltage at resonance?

4. How does increasing R affect the sharpness and height of the resonance curve?

5. Compare series and parallel LCR resonance — which one gives minimum impedance and which gives maximum impedance at resonance?

6. Explain how this resonance principle is used in radio-frequency tuning circuits.

14. Maxwell's Equations & Plane Electromagnetic Wave Solutions

Aim

To explore how Maxwell's four equations, together, predict self-sustaining transverse electromagnetic waves in free space, and to verify the relations between E, B, their phase, and the speed of light.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$c = 1/\sqrt{\mu_0 \varepsilon_0}$, $E_0/B_0 = c$, E ⊥ B ⊥ direction of propagation

c = speed of light in vacuum (m/s)
μ_0 = permeability of free space
ε_0 = permittivity of free space
E_0, B_0 = peak amplitudes of the electric and magnetic fields

Labelled diagram of the experimental apparatus:

Observation Table

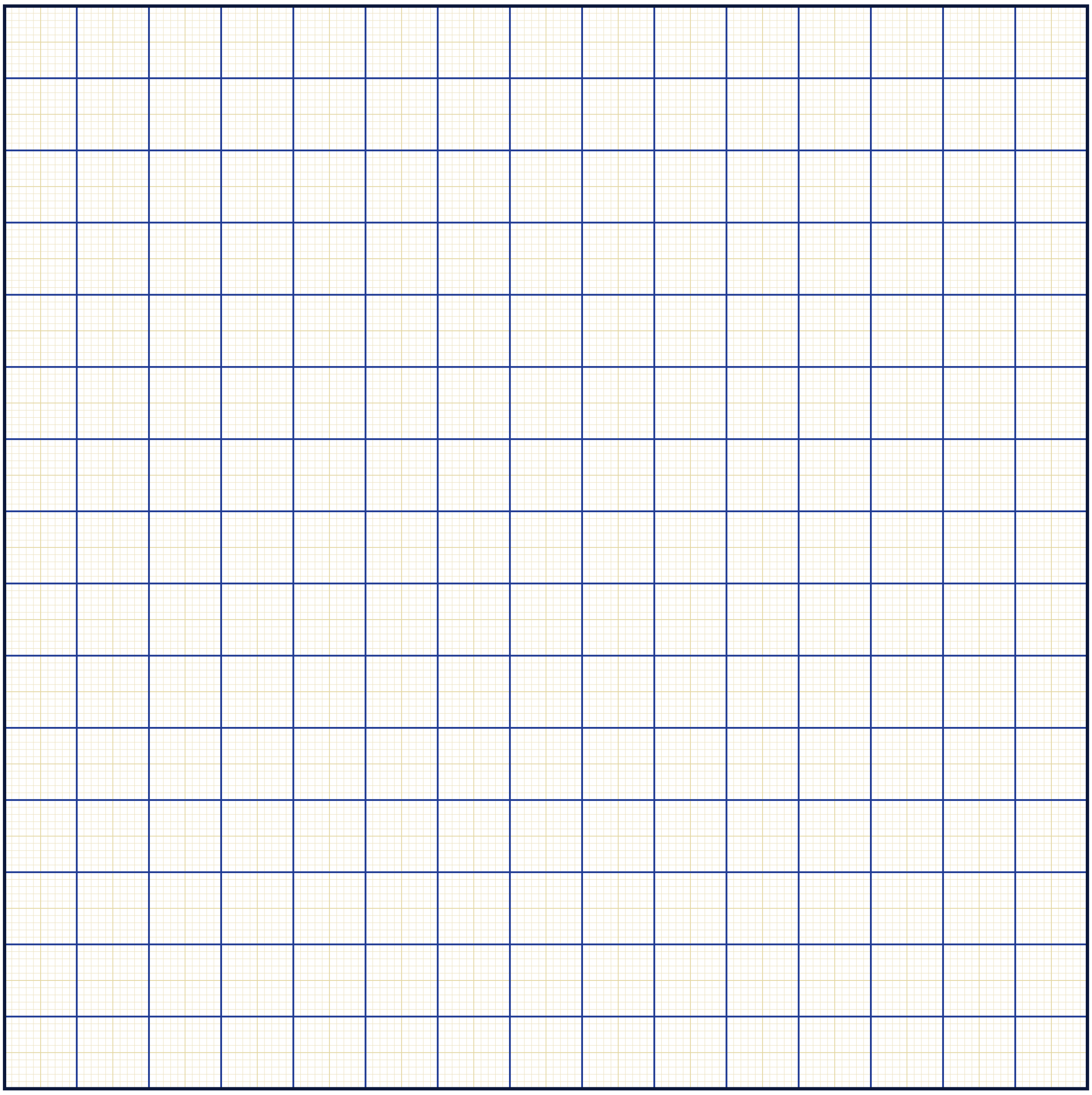
S. No.	Frequency f (MHz)	E_0 (V/m)	Computed B_0 (nT)	Wavelength λ (m)	E_0/B_0 ratio ($\times 10^8$ m/s)

S. No.	Frequency f (MHz)	E_0 (V/m)	Computed B_0 (nT)	Wavelength λ (m)	E_0/B_0 ratio ($\times 10^8$ m/s)

Graph

E and B Field Snapshot vs Position

Field (normalized)



z (m)

Calculation

Result

Quantity	Measured value	Theoretical value (c)	% deviation

Maximum Permissible Error

Formula: $\Delta B_0/B_0 = \Delta E_0/E_0$

Since $B_0 = E_0/c$ with c an exact constant, the fractional error in B_0 equals the fractional error in the measured E_0.

Quantity	Measured Value	Least Count (LC)
E■ — E amplitude (V/m)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Starting from Maxwell's curl equations, outline the steps to derive the electromagnetic wave equation.

2. Why must E and B be mutually perpendicular and both transverse to the direction of propagation?

3. Show dimensionally that $1/\sqrt{\mu_0\varepsilon_0}$ has the units of speed.

4. What physical role does Maxwell's displacement current term play in predicting wave propagation?

5. How is the Poynting vector defined, and what does it represent physically?

6. Why was the prediction of electromagnetic waves by Maxwell considered one of the great unifications in physics?

15. Propagation of EM Waves in Dielectric & Conducting Media (Skin Depth)

Aim

To study how the speed, wavelength, and attenuation of an electromagnetic wave change on entering a dielectric or a conducting medium, and to determine the skin depth for a conductor as a function of frequency and conductivity.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

Dielectric: $v = c/\sqrt{\mu_r \varepsilon_r}$ | Conductor: $\delta = \sqrt{2/\omega\mu\sigma}$

v = wave speed inside the dielectric (m/s)
$n = \sqrt{\mu_r \varepsilon_r}$ = refractive index
δ = skin depth in a conductor (m)
ω = angular frequency (rad/s)
σ = conductivity (S/m)

Labelled diagram of the experimental apparatus:

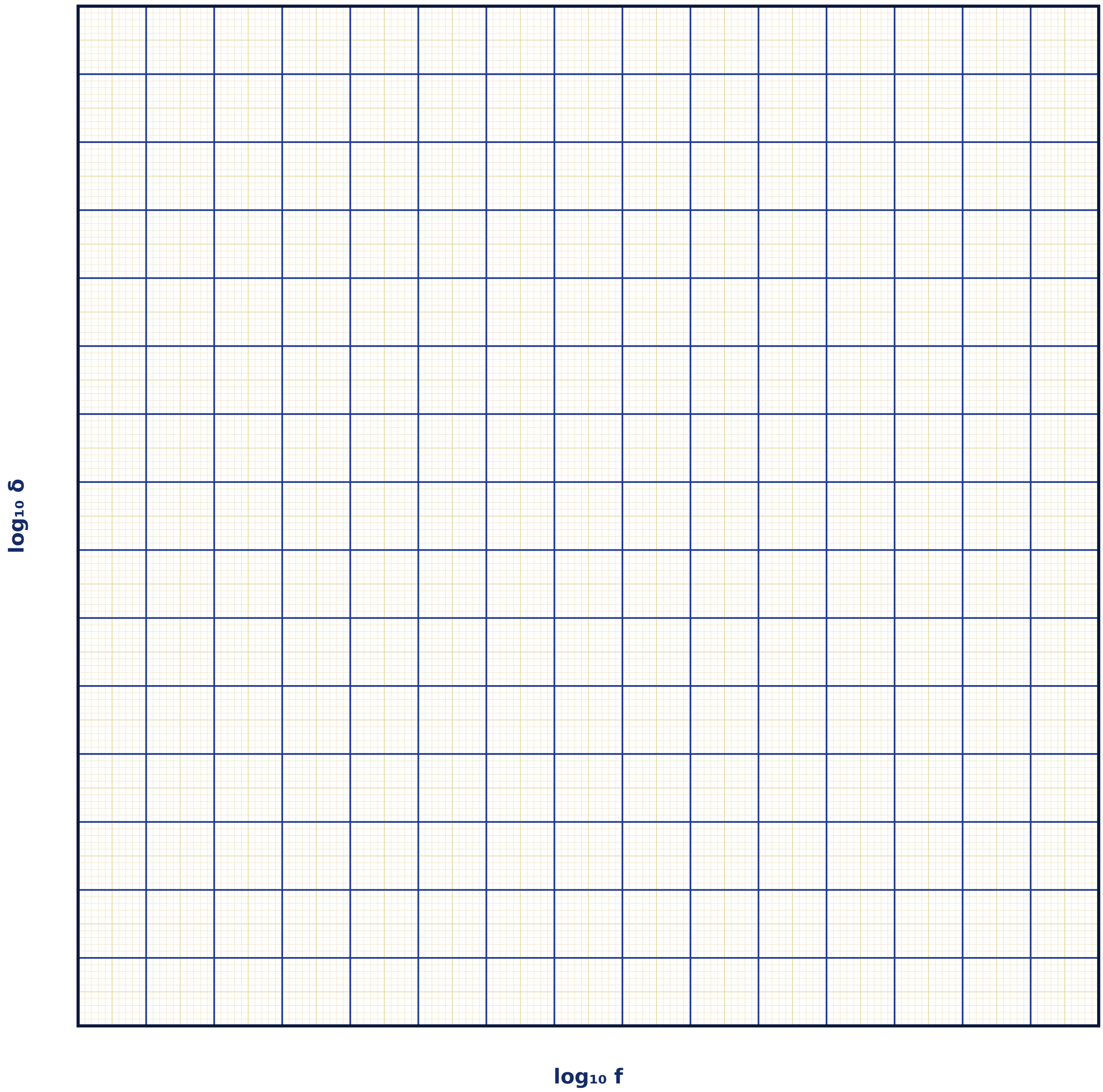
Observation Table

S. No.	Mode	Frequency f (MHz)	ε_r or σ	Wave speed / Skin depth	Wavelength (dielectric mode)

S. No.	Mode	Frequency f (MHz)	εr or σ	Wave speed / Skin depth	Wavelength (dielectric mode)

Graph

Skin Depth vs Frequency (log-log)



Calculation

Result

Case	Measured quantity	Theoretical value	% deviation

Maximum Permissible Error

Formula: $\Delta\delta/\delta = \frac{1}{2}(\Delta f/f + \Delta\sigma/\sigma)$

$\delta \propto 1/\sqrt{f\sigma}$, so the fractional errors in f and σ are each halved before summing.

Quantity	Measured Value	Least Count (LC)
f — Frequency (MHz)		
σ — Conductivity (MS/m)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive the refractive index $n = \sqrt{\mu_r \varepsilon_r}$ from the wave equation in a dielectric medium.

2. Why does an EM wave attenuate exponentially inside a conductor but not inside an ideal (lossless) dielectric?

3. Derive the expression for skin depth starting from the wave equation with a conduction-current term.

4. Why are radio antennas and high-frequency circuit traces often silver-plated even though bulk silver is only marginally more conductive than copper?

5. How does skin depth explain why AC power transmission cables sometimes use stranded or hollow conductors?

6. Relate the concept of skin depth to electromagnetic shielding effectiveness of an enclosure.

16. Reflection, Refraction & Brewster's Angle for EM Waves

Aim

To study the reflection and transmission of an electromagnetic wave at a dielectric interface using the Fresnel equations, and to locate Brewster's angle at which p-polarized reflection vanishes.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

Snell's law: $n_1 \sin\theta_i = n_2 \sin\theta_t$ | Brewster's angle: $\theta_B = \arctan(n_2/n_1)$

n_1, n_2 = refractive indices of the incident and transmitting media
θ_i, θ_t = angles of incidence and refraction
θ_B = Brewster's angle, at which p-polarized reflectance is zero

Labelled diagram of the experimental apparatus:

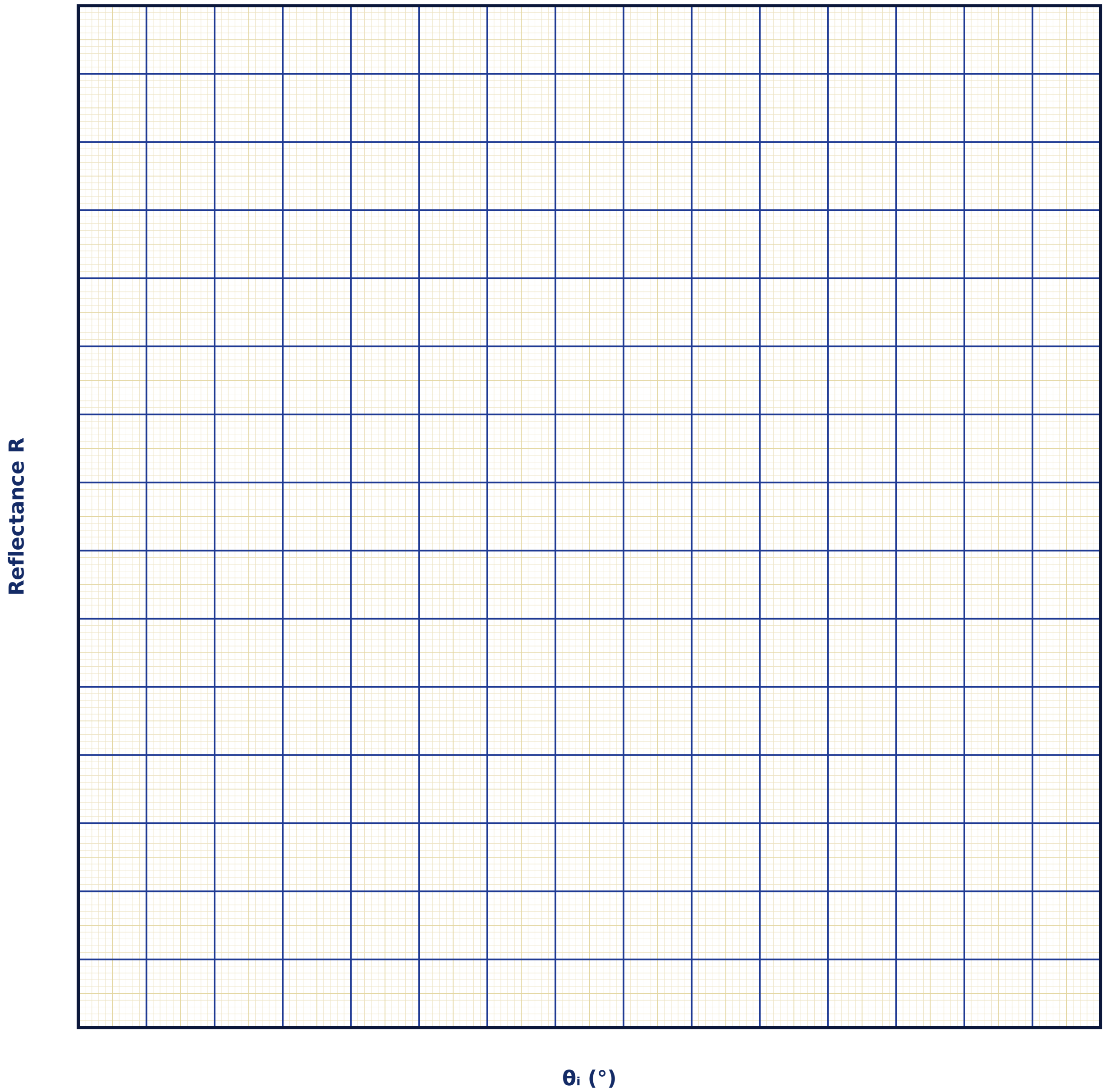
Observation Table

S. No.	n_1	n_2	θ_i (°)	R_p (%)	R_s (%)

S. No.	n_1	n_2	θ_i (°)	R_p (%)	R_s (%)

Graph

Reflectance vs Angle of Incidence



Calculation

Result

n_2/n_1	Brewster angle (measured)	Brewster angle (arctan n_2/n_1)	% deviation

Maximum Permissible Error

Formula: $\Delta\theta_B \approx [1/(1+(n_2/n_1)^2)]\cdot(n_2/n_1)\cdot(\Delta n_2/n_2 + \Delta n_1/n_1)$

Obtained by differentiating $\theta_B = \arctan(n_2/n_1)$ with respect to the ratio n_2/n_1.

Quantity	Measured Value	Least Count (LC)
n■ — n1		
n■ — n2		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Derive Brewster's angle condition from the Fresnel equation for p-polarized reflectance.

2. Why does no Brewster angle exist for s-polarized light?

3. Show that at Brewster's angle the reflected and refracted rays are perpendicular.

4. How are Brewster windows used inside a laser cavity, and why?

5. Distinguish Brewster's angle from the critical angle for total internal reflection.

6. How do polarizing sunglasses exploit Brewster-angle reflection from horizontal surfaces like water or roads?

17. Dielectric Polarization & the Clausius–Mossotti Relation

Aim

To study how an external electric field polarizes a dielectric material, relate the macroscopic polarization to the microscopic atomic polarizability using the Clausius–Mossotti relation, and observe the resulting bound charges.

Date performed: ___________________   Simulator panel used: ____________________________

Formula Used

$$P = \varepsilon_0(\varepsilon_r - 1)E, \quad (\varepsilon_r - 1)/(\varepsilon_r + 2) = N\alpha/(3\varepsilon_0)$$

P = macroscopic polarization (C/m^2)
εr = relative permittivity (dielectric constant)
E = applied macroscopic field (V/m)
N = number density of molecules (m^{-3})
α = molecular polarizability ($C \cdot m^2/V$)

Labelled diagram of the experimental apparatus:

Observation Table

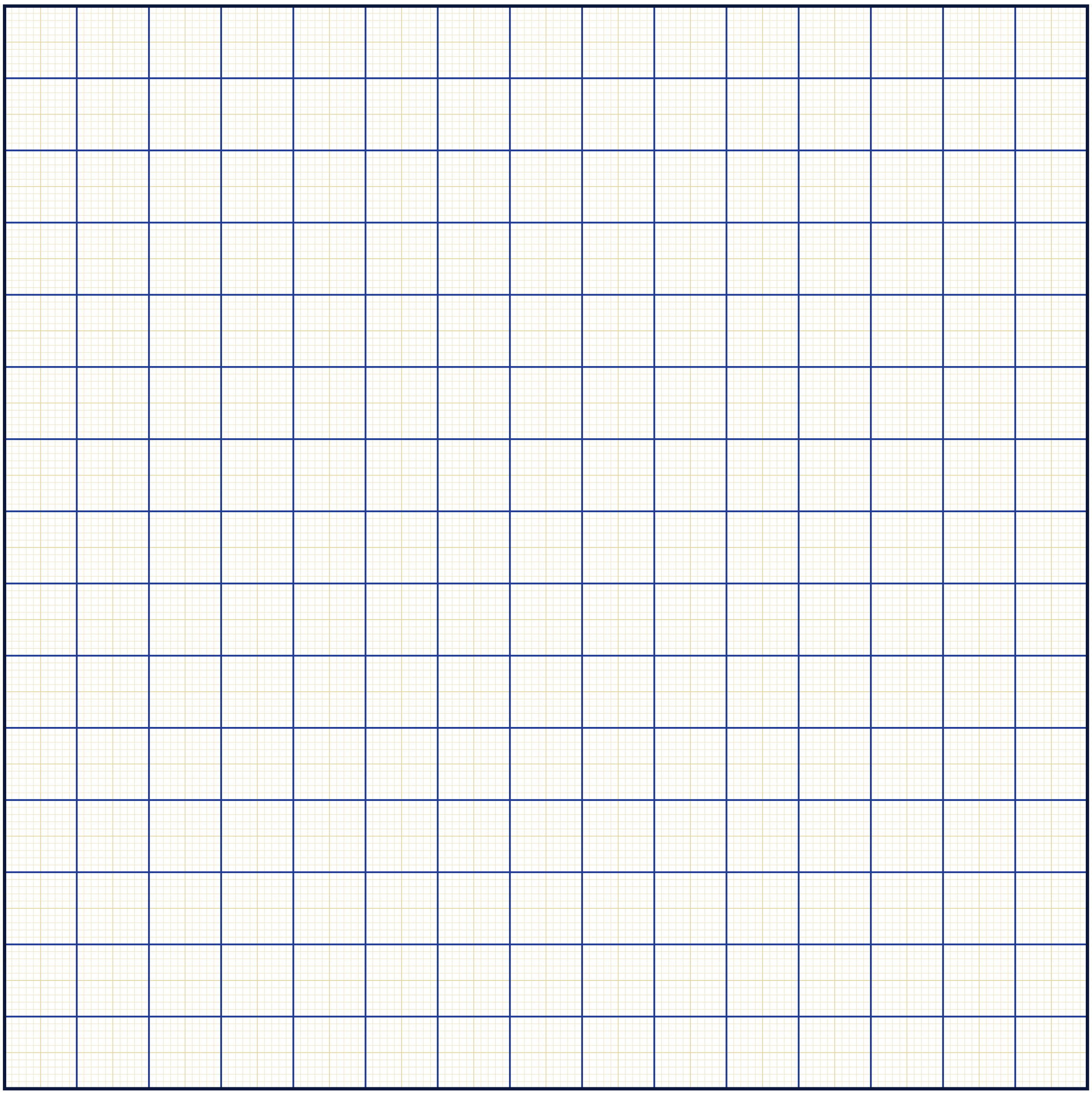
S. No.	N ($\times 10^{28}$ m^{-3})	α ($\times 10^{-40}$ $C \cdot m^2/V$)	εr (Clausius–Mossotti)	Applied field E (kV/m)	Polarization P (μC/m²)

S. No.	N ($\times 10^{28}$ m^{-3})	α ($\times 10^{-40}$ $C\cdot m^2/V$)	εr (Clausius–Mossotti)	Applied field E (kV/m)	Polarization P (μC/m^2)

Graph

Dielectric Constant vs Number Density N

εr



N ($\times 10^{28}$ m^{-3})

Calculation

Result

Case	εr computed	Bound charge σ_b	Remarks

Maximum Permissible Error

Formula: via Clausius–Mossotti; Δεr estimated numerically from ΔN/N + Δα/α

εr is a nonlinear function of Nα, so its propagated error is evaluated numerically from the fractional errors of N and α rather than a simple closed form.

Quantity	Measured Value	Least Count (LC)
N — Number density ($\times 10^{2}$■ m■^{3})		
α — Polarizability (×10■■■ C·m^{2}/V)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Define polarization P and relate it to the electric susceptibility χ_e.

2. Derive the Clausius–Mossotti relation, explaining the role of the local (Lorentz) field correction.

3. What is meant by 'bound charge', and where does it appear on/in a polarized dielectric?

4. Why does the simple relation $\varepsilon_r - 1 \approx N\alpha/\varepsilon_0$ (without the local-field correction) become inaccurate at high densities?

5. How does the Clausius–Mossotti relation connect to the Lorentz–Lorenz relation for the refractive index?

6. Explain qualitatively what a 'polarization catastrophe' represents physically (link to ferroelectricity).

18. Magnetic Hysteresis & the B–H Curve of Ferromagnetic Materials

Aim

To trace the B–H hysteresis loop of a ferromagnetic material as the applied field H is cycled, and to determine the saturation magnetization, retentivity (remanence), coercivity, and hysteresis energy loss per cycle.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$H = nI$, Hysteresis loss per cycle per unit volume = $\oint H\, dB$ (loop area)

H = magnetizing field (A/m)
B = magnetic flux density in the material (T)
B_r = retentivity / remanence (T)
H_c = coercivity (A/m)
B_sat = saturation flux density (T)

Labelled diagram of the experimental apparatus:

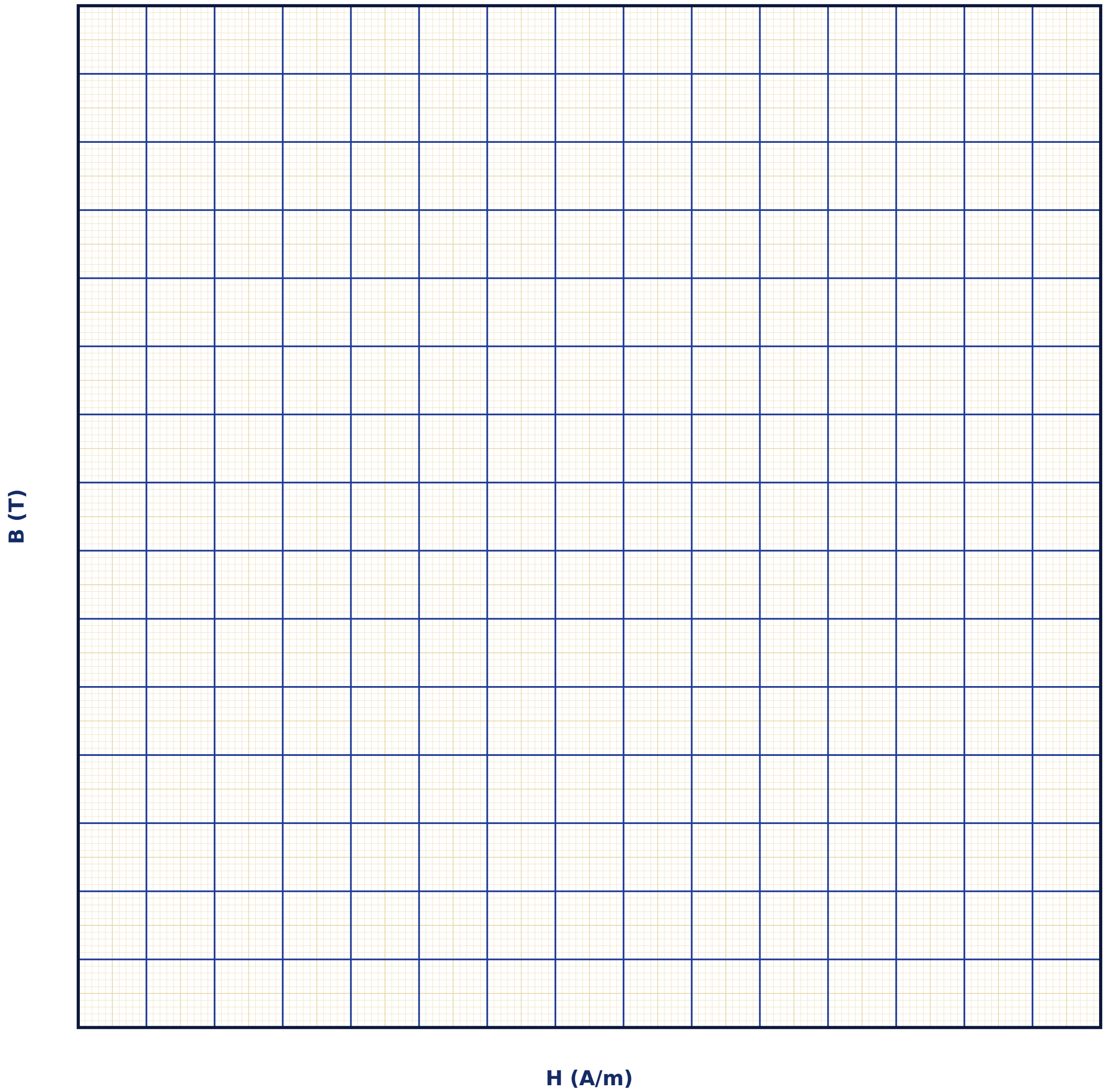
Observation Table

S. No.	Material	H (A/m)	B (T)	Cycle number

S. No.	Material	H (A/m)	B (T)	Cycle number

Graph

B-H Hysteresis Loop



Calculation

Result

Material	B_sat (T)	Retentivity B_r (T)	Coercivity H_c (A/m)	Loop area / cycle loss

Maximum Permissible Error

Formula: $H_c \approx 0.15 \times H_{max}$ (model estimate); $\Delta H_c/H_c = \Delta H_{max}/H_{max}$

In this model, coercivity is estimated as a fixed fraction of the peak applied field; its uncertainty tracks the peak-field setting's fractional uncertainty.

Quantity	Measured Value	Least Count (LC)
H_{max} — Peak applied field (A/m)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Define retentivity and coercivity, and explain how each is read off a B–H loop.

2. Why does the area enclosed by the hysteresis loop represent energy dissipated as heat?

3. Distinguish between magnetically 'soft' and 'hard' materials in terms of their hysteresis loops, and give one application of each.

4. Why do transformer cores use soft magnetic materials with narrow loops rather than hard materials?

5. What is a domain, and how does domain-wall motion explain the irreversibility seen in hysteresis?

6. How does the hysteresis loss per cycle relate to the total power loss in an AC-excited transformer core operating at 50 Hz?